

RESPONSES

General responses

Response: Thank you to the editor and the reviewers for their continued valuable input.

Editor

Response: After a recent conversation with Paulo Guimarães, I realized that his team has been moving in a similar direction. I have distinguished this proposal from theirs in a footnote:

Closely related to the proposal here is the mechanism used in Guimarães (2023). However, they continue to host the validating infrastructure on the agency's servers, in a centralized fashion, thus not leveraging the full cost savings as the decentralized or outsourced method proposed here.

Referee 2

This is a very important topic that many data scientists will find interesting. I recommend publication. I believe it would improve accessibility if more time was spent in the abstract and introduction linking containers to the regular work of data scientist: for example, why is this article being written? what prompted it? what does the author hope the reader will get out of it? What is the connection to Census (this is central and never clearly explained)?

Response:

Thank you for framing the questions in this way to broaden the audience for the article. I have added or expanded some sentences to the introduction to reference the current alternatives that are "costly to maintain" and require more effort from the researchers.

As an alternative, data custodians provide confidential data to researchers in restricted-access environments, often still requiring physical presence in expensive to maintain secure computer rooms managed by the data custodian or universities (Vilhuber, 2024b; Weinberg et al., 2007).

This article will set out to combine various lessons learned from both the literature and past experiments in a variety of domains to propose an alternative or complementary approach to providing access to confidential data via synthetic data or physical access, reducing the effort required by researchers to access these data.

The connection to the Census is not central at all to the arguments here. It is used as a motivating example, because of the author's prior experience with the SSB on the Synthetic Data Server. But the mechanism is a general one. The Census Bureau is, of course, central to the illustrative use case (starting on page 11). To avoid the impression that this is primarily about the Census Bureau, I chosen to de-emphasize the Census Bureau (and Codeocean) in the example:

Without loss of generality, the use case is specifically chosen to extend the type of analysis previously conducted on the SDS with SSB data, relying on validation by the U.S. Census Bureau.

and in some of the subsequent sentences:

In this configuration, the data owner (here: Census Bureau) could allow the container system provider (here: Codeocean) to house both the synthetic dataset as well as...

A question that many readers will have pertains to the recommendation to use commercial vendors, ie. Code Ocean. With vendors there is always the risk they vanish, or change policies, etc. How does the author propose the research community mitigates this risk, while still enabling access to and validation of confidential data? Readers will remember when this exact thing happened to docker / docker hub a few years ago which was widely used as a free service for container hosting, like Code Ocean today, then became pay to play with a policy change.

Response: I had partially addressed the fact that vendors can go away in a footnote in the text. I have moved and elevated this footnote into a full paragraph in a section on "Caveats". I note that some vendors have disappeared, but that this is fundamentally not the same problem. The use of any provider is primarily a transient process, destined to produce the final analysis in the confines of the statistical agency. It is the final output that primarily needs preservation, not the intermediate output produced via a platform. That being said, as the referee's next comment points out, the reproducible package generated with the low-quality synthetic data has value, as a more accessible replication package. I discuss this, without expanding too much.

Finally, some literature was missed that pioneered the idea of releasing smaller / reduced datasets for the purposes of reproducibility and access. Scientific Tests and Continuous Integration Strategies to Enhance Reproducibility in the Scientific Software Context Authors: Matthew Krafczyk, August Shi, Adhithya Bhaskar, Darko Marinov, Victoria Stodden P-RECS '19: Proceedings of the 2nd International Workshop on Practical Reproducible Evaluation of Computer Systems Pages 23 - 28

Response: Thank you for pointing out the analogy. I have added a footnote in Section "A Proposal using Containers" that includes this reference, and also notes that the provision of fake data for the same purpose has been policy in some management and economics journals since around 2019. Interestingly, it is not policy in the journals that I am Data Editor, despite my attempts to convince the editorial boards to require this. I do note that the novelty only pertains to "purposes of reproducibility". The whole point of synthetic data has been from the outset (Rubin, 1993) to provide access to otherwise confidential data.

Referee 3

Previous comment: page 5 middle paragraph: The second sentence starts with talking about Statistics Canada and microdata.no but ends with the 2018 paper. My previous comment, now is about page 8 middle paragraph, tried to say that the way the sentence is structured made it sound like Statistics Canada and microdata.no also use the method from the 2018 paper. If that is not the case, then the sentence would need some modification.

Response: Apologies for missing that. I agree with the point, and have split the sentence into two, for greater clarity:

The aforementioned Statistics Canada and microdata.no systems strongly limit the type of analyses that are feasible by restricting the software keywords that can be used (RTRA) or by creating a structured new statistical language (microdata.no). The system proposed in Barrientos et al. (2018) limits the types of analysis that can be run and validated.

It would be helpful to include an equivalent R example.

Response: A R example will be added on the Github repository associated with the example, as doing so within this article would make it too long. The important point is, in fact, that the system proposed can support many programming languages, as long as the disclosure mechanism of the agency can be applied. I have added an appropriate "Without loss of generality" when introducing the concrete example. I do note that for the particular use case, the data custodian (Census Bureau) does not actually allow for R code as of this writing.